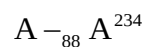
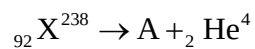


Solutions

Structure of Atom (Subjective Assignment)

1.



$$\text{No. of Neutron} = 234 - 88 = 146$$

2.

$$E = \frac{hc}{\lambda}$$

$$\frac{E_1}{E_2} = \frac{\lambda_1}{\lambda_2}$$

$$E_1.E_2 = 1:4$$

3.

$$\bar{\nu} = 1200 \text{ KHZ}$$

$$\bar{\nu} = ?$$

$$\bar{\nu} = \frac{1}{\lambda}$$

$$\bar{\nu} = \frac{\nu}{c}$$

$$\bar{\nu} = \frac{1200 \times 10^3}{3 \times 10^8}$$

$$= 4 \text{ km}^{-1}$$

Hence in one Km there are 4 waves.

4.

$$\text{Energy of A} \rightarrow \frac{hc}{8000} \dots (i)$$

$$\text{Energy of B} \rightarrow \frac{hc}{4000} \dots(\text{ii})$$

$$P = \frac{E}{t}$$

$$\frac{P_1}{P_2} = \frac{E_1}{E_2}$$

$$\frac{100}{P_2} = \frac{4000}{8000} \Rightarrow P_2 = 200$$

5.

$$\lambda = 230 \text{ n.m.}$$

$$\lambda = 180 \times 10^{-9} \text{ m}$$

$$\lambda_0 = 230 \times 10^{-9} \text{ m}$$

$$\frac{hc}{\lambda} = \frac{hc}{\lambda_0} + \text{KE}$$

$$\text{KE} = \frac{hc}{\lambda} - \frac{hc}{\lambda_0}$$

$$\text{KE} = hc \left[\frac{\lambda_0 - \lambda}{\lambda \times \lambda_0} \right]$$

$$\text{KE} = 6.64 \times 10^{-34} \times 3 \times 10^8 \left[\frac{230 \times 10^{-9} - 180 \times 10^{-9}}{180 \times 10^{-9} - 230 \times 10^{-9}} \right]$$

$$\text{KE} = 6.64 \times 3 \times 10^{-26} \left[\frac{50 \times 10^{-9}}{41,400 \times 10^{-1.8}} \right]$$

$$\text{KE} = 0.024 \times 10^{-17}$$

$$\text{KE} = 2.4 \times 10^{-19} \text{ J}$$

6.

$$\frac{hc}{\lambda} = W + \text{KE}$$

$$\frac{6.64 \times 10^{-34} \times 3 \times 10^8}{4 \times 10^{-7}} = 2.13 \times 10^{-19} \times 1.6 \times \text{KE}$$

$$= 4.98 \times 10^{-19} - 2.13 \times 10^{-19} \times 1.6 - \text{KE}$$

$$\text{KE} \approx 1.572 \times 10^{-19}$$

$$\frac{1}{2} m v^2 = 1.572 \times 10^{-19}$$

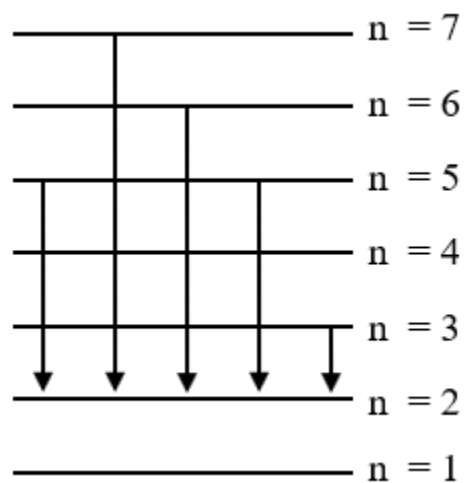
$$v^2 = \frac{2 \times 1.572 \times 10^{-19}}{9.31 \times 10^{-31}}$$

$$v^2 = 0.33 \times 10^{12}$$

$$v = 0.19 \times 10^6$$

$$v = 1.9 \times 10^5 \text{ m/sec}$$

7.



$$\left. \begin{array}{l} 7 \rightarrow 2 \\ 6 \rightarrow 2 \\ 5 \rightarrow 2 \\ 4 \rightarrow 2 \\ 3 \rightarrow 2 \end{array} \right\} \text{Five Balmer Lines}$$

8.

Paschan $\rightarrow$ means

$n \rightarrow 3$ Transition

$$\Rightarrow 2.18 \times 10^{-18} \times \frac{Z^2}{n^2}$$

$$\Rightarrow \frac{2.18 \times 10^{-18}}{n^2} - \frac{2.18 \times 10^{-18}}{9} = 18.16 \times 10^{-20}$$

$$2.18 \times 10^{-18} \left(\frac{1}{n^2} - \frac{1}{9} \right) = 18.16 \times 10^{-20}$$

$$\left(\frac{1}{n^2} - \frac{1}{9} \right) = 8.33 \times 10^{-2}$$

$$\frac{1}{n^2} = 0.11 + 8.33 \times 10^{-2}$$

$$\frac{1}{n^2} = 0.194$$

$$n = 7$$

(Then Quantum No is 7)

9.

$$mvr = \frac{nh}{2\pi}$$

If $n = 2$

Then Angular momentum is $\frac{h}{\pi}$

10.

$$n \rightarrow 4, l = 1 \rightarrow 4p$$

$$n \rightarrow 4, l = 0 \rightarrow 4s$$

$$n \rightarrow 3, l = 2 \rightarrow 3d$$

$$n \rightarrow 3, l = 1 \rightarrow 3p$$

applying $(n + l)$ rule

$$2p > 4s > 3d > 4p$$

11.

$$\Delta P \cdot \Delta x \geq \frac{h}{4\pi}$$

$$m \cdot \Delta V \cdot \Delta x \geq \frac{h}{4\pi}$$

$$10^{-3} \times \frac{3.313}{\pi} \times 10^{-3} \times \Delta x \geq \frac{6.64 \times 10^{-34}}{4\pi}$$

$$\Delta x = 2 \times 10^{-25} \text{ m}$$

12.

$$2\pi r = n \times \lambda$$

$$\lambda = \frac{2\pi r}{2}$$

$$\lambda = \pi r \text{ which is } 2 \text{ n.m}$$

13.

$$\Delta x = 10^{-10}$$

$$\Delta V = 5.27 \times 10^{-24} \text{ m / sec}$$

$$m \cdot \Delta V \cdot \Delta x = \frac{h}{4\pi}$$

$$m = \frac{6.64 \times 10^{-34}}{4 \times 3.14 \times 5.27 \times 10^{-24} \times 10^{-10}}$$

$$m = 0.1003 \text{ kg}$$

14.

$$\Rightarrow -\frac{E_0}{16} + \frac{E_0}{9} = x$$

$$\frac{7E_0}{16 \times 9} = x$$

$$E_0 = \frac{16 \times 9 \times x}{7}$$

$$\text{Then } \frac{-E_0}{9} + \frac{-E_0}{4}$$

$$E_0 \left(\frac{5}{36} \right)$$

$$= \frac{20x}{7}$$

15.

K, L, M, N

2, 8, 8, 2

→ According to given condition

Then It's Atomic No is 20.

16.

$A^+ \rightarrow 2, 8, 18$

Then Atomic No. is $\Rightarrow 29$

17.

$$(A + 0.25) = \frac{A \times P_1 + (A + 2) \times P_2}{100}$$

$$(A + 0.25) \times 100 = AP_1 + (A + 2)P_2$$

$$(A + 0.25) \times 100 = A \times (100 - P_2 + AP_2 + 2P_2)$$

$$A + 100 + 25 = A \times 100 - AP_2 + AP_2 + 2P_2$$

$$P_2 = \frac{25}{2}$$

$$P_1 = 87.5$$

18.

$$r = \frac{r_0 \times n^2}{Z}$$

$$r = 0.529 \times n^2 A_0$$

$$\text{Then } 0.132 n^2 = \frac{0.529 \times n^2}{Z}$$

$$Z = 4$$

Element is Be.

19.

$$\frac{hc}{\lambda} = R \left[\frac{1}{9} - \frac{1}{\infty} \right] \times hc$$

Different in energy

$$\frac{1096707}{10^{-7}} \times \left[\frac{5}{36} \right] \times hc = h\nu$$

$$\nu = \frac{1096707 \times 5 \times 3 \times 10^8}{36 \times 10^{-2}}$$

$$\nu = 45698 \times 10^{10} \text{ Hz}$$

20.

$$mvr = \frac{nh}{2\pi}$$

Difference in angular momentum

$$\Rightarrow (4.20 \times 1.05) \times 10^{-34} = 3.15 \times 10^{-34}$$

$$\Rightarrow 3.15 \times 10^{-34} \left(n_2 - n_1 \times \frac{h}{2\pi} \right)$$

$$(n_2 - n_1) = 3$$

Then transition may be

$$\left[\begin{array}{l} 4 \rightarrow 1 \\ 5 \rightarrow 2 \\ 6 \rightarrow 3 \end{array} \right] \text{ Etc.}$$

21.

Lighter particle will give the proper deflection hence we use ${}_{x-1}^{4-2}\text{B}^{+1}$